A five-strip covering certificate for planar null geodesics in an equal-mass Majumdar–Papapetrou spacetime

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Abstract

We give explicit covering certificates for a first-return map of planar null geodesics in an equal-mass Majumdar–Papapetrou spacetime. Both masses are one and their coordinate separation is two. Five disjoint strips in three affine section boxes satisfy entry and exit inequalities for the first positive upward return to $z = 0$. Two interval implementations, one using binary64 Taylor enclosures and the other CAPD with 160-bit MPFR intervals, validate the same five coverings. They share the specified model, boxes and strips; their agreement does not independently validate that shared input. The resulting edge shift has spectral radius $\varphi = (1 + \sqrt{5})/2$. Hence a compact invariant set of the return map has topological entropy at least $\log \varphi$. The contribution is a checkable, parameter-specific certificate and a bound per return; it is not a claim of the first rigorous proof of chaos in a two-centre spacetime.

1 Statement and scope

Planar photon scattering in two-centre Majumdar–Papapetrou geometries has a substantial symbolic and numerical literature. Contopoulos [1] studied photon and particle orbits around two fixed extreme Reissner–Nordström black holes, relating capture and escape to unstable periodic orbits and Cantor-like initial-condition sets. Dettmann, Frankel and Cornish [2] used numerical Lyapunov exponents and fractal basin boundaries to characterize chaotic motion in multi-black-hole spacetimes. Cornish and Gibbons [3] describe symbolic dynamics in a family of two-centre models, and Shipley and Dolan [4] develop a detailed account of null scattering and black-hole shadows. Our object is a specified first-return map, with rational section data and interval certificates for whole sets of initial conditions.

Qualitative positive entropy for the continuous-time optical geodesic flow on a compact invariant set also follows from the compact-core case of Bolotin and Kozlov's Theorem 2.1 [5]. Appendix A gives our application to the optical metric: the specified compact domain has Euler characteristic -1 and three strictly geodesically convex boundary components. This general result does not specify the present strips or a bound per return. Our method follows the established covering-relation approach [6], including its computer-assisted applications to the Hénon, Rössler and Lorenz systems listed in its introduction; the covering method itself is not new. Entropy values with different word lengths or time parametrizations are not treated as the same quantity.

The finite data below specify the claim completely. Write N_O, N_A, N_B for the affine boxes in Section 3, and $K_{ij} \subset N_i$ for the five strips in Table 3.

Theorem 1. *For the equal-mass system and section defined in Section 2, the first-return map P, whose domain requires return before departure from* $\operatorname{int} \mathcal{D}$*, is defined and continuous on each closed K_{ij}. All return segments remain in the interior of*

$$\mathcal{D} = \{r_+ \geq 1/5,\ r_- \geq 1/5,\ x^2 + z^2 \leq 64\}.$$

With one exit coordinate and one entry coordinate, the following covering relations hold:

$$K_{OO} \overset{P}{\Longrightarrow} N_O, \quad K_{OA} \overset{P}{\Longrightarrow} N_A, \quad K_{AB} \overset{P}{\Longrightarrow} N_B, \quad K_{BO} \overset{P}{\Longrightarrow} N_O, \quad K_{BA} \overset{P}{\Longrightarrow} N_A. \tag{1}$$

Each K_{ij} also covers every listed successor strip $K_{jk} \subset N_j$, so every transition of the matrix in (8) is realized by a covering relation. There is a nonempty compact set $\Lambda \subset K_{OO} \cup K_{OA} \cup K_{AB} \cup K_{BO} \cup K_{BA}$ with $P(\Lambda) = \Lambda$ such that

$$h_{\mathrm{top}}(P|_\Lambda) \geq \log \frac{1+\sqrt{5}}{2} \geq 0.481211825059603. \tag{2}$$

The theorem is computer-assisted: its finite inequalities are established by the two interval implementations described in Section 5. The topological deduction is a prose proof using covering relations [6]; no Lean formalization is supplied. The statement concerns these parameters and this return map. It does not assert hyperbolicity, conjugacy, a description of the full shadow, or a lower bound for a time-one flow map.

2 Model and first-return domain

Use signed Cartesian coordinates (x, y, z), with the two centres at $(0, 0, \pm 1)$. On the invariant plane $y = p_y = 0$, set

$$r_\pm = \sqrt{x^2 + (z \mp 1)^2}, \qquad \mathcal{U} = 1 + \frac{1}{r_+} + \frac{1}{r_-}. \tag{3}$$

The metric is $ds^2 = -\mathcal{U}^{-2}dt^2 + \mathcal{U}^2(dx^2 + dy^2 + dz^2)$. Fix $p_t = -1$ and the null constraint

$$H = \tfrac{1}{2}\big[-\mathcal{U}^2 + (p_x^2 + p_z^2)/\mathcal{U}^2\big] = 0.$$

This fixes the affine-time scale. In the null reduction, put $v = (v_x, v_z) = (p_x, p_z)/\mathcal{U}^2$ and $Q = (x, z)$. The equations used in both certificates are

$$Q' = v, \qquad v' = \frac{2}{\mathcal{U}}\big(\nabla\mathcal{U} - v(\nabla\mathcal{U} \cdot v)\big), \qquad \frac{dt}{d\lambda} = \mathcal{U}^2, \tag{4}$$

where prime denotes differentiation with respect to λ. These agree with the planar null Hamilton equations in [4, Section II.1]. The identity

$$(|v|^2)' = \frac{4}{\mathcal{U}}(\nabla\mathcal{U} \cdot v)(1 - |v|^2) \tag{5}$$

preserves $|v| = 1$. Equation (4), regarded as a smooth extension off this hypersurface, is also used to propagate outer interval sets.

Parameterize the upward section by

$$\iota(q, w) = (q, 0, w, \sqrt{1 - w^2}), \qquad |w| < 1. \tag{6}$$

For a section point whose solution stays in $\operatorname{int}\mathcal{D}$ until its next upward crossing, let $\tau(q, w) > 0$ be its first such crossing time and define

$$P(q, w) = \pi\Phi_{\tau(q,w)}(\iota(q, w)), \qquad \pi(x, z, v_x, v_z) = (x, v_x).$$

The initial zero at $\lambda = 0$ is excluded. The domain of P consists of points for which this finite first return exists before departure from $\operatorname{int}\mathcal{D}$. The certificates establish this property on all five closed strips, including their boundaries. The artificial boundary of $\mathcal{D}$ is a validation boundary, not an event horizon.

3 Exact boxes and strips

Let $c_i \in \mathbb{R}^2$, $e_i^u, e_i^s \in \mathbb{R}^2$ and $r_i^u, r_i^s > 0$ be the data in Tables 1–2. Every displayed finite decimal in these two tables is an exact rational number. Define

$$T_i = [\, r_i^u e_i^u \;\; r_i^s e_i^s \,], \qquad \psi_i(u,s) = c_i + T_i(u,s)^{\mathsf{T}}, \qquad N_i = \psi_i([-1,1]^2). \tag{7}$$

The superscripts identify the exit and entry coordinates; no eigenvector or hyperbolicity assertion is implicit in the notation.

i	$c_{i,q}$	$c_{i,w}$	r_i^u	r_i^s
O	2.264907835107	0	0.00015	0.0015
A	2.251945728135	−0.003640514957	0.000005	0.0015
B	2.089732323743	0.058272583543	0.00015	0.003

Table 1: Exact centres and radii in the section coordinates (q, w).

i	e_i^u	e_i^s
O	(0.9521671202, 0.3055777729)	(0.9521671202, −0.3055777729)
A	(0.9516691518, 0.3071250974)	(0.9514626094, −0.3077643628)
B	(0.9424498218, 0.3343476236)	(0.9453262599, −0.3261261449)

Table 2: Exact frame vectors. They are used as given, without normalization.

For an edge ij in Table 3, write its listed interval as $[a_{ij}, b_{ij}]$ and set

$$K_{ij} = \psi_i([a_{ij}, b_{ij}] \times [-1,1]).$$

In its own h-set chart, the exit coordinate is rescaled from $[a_{ij}, b_{ij}]$ to $[-1,1]$; the entry coordinate is unchanged. The frame determinants are nonzero. Exact rational geometry checks give positive gaps between the q-projections of N_O, N_A, N_B:

$$\begin{aligned} \operatorname{gap}(N_O, N_A) &= \frac{9959078963811}{10^{15}}, \\ \operatorname{gap}(N_O, N_B) &= \frac{1706270893627}{10^{13}}, \\ \operatorname{gap}(N_A, N_B) &= \frac{157804105879171}{10^{15}}. \end{aligned}$$

The two strips within N_O, and the two within N_B, have separated exit intervals. Consequently all five K_{ij} are pairwise disjoint.

Strip	Exit interval in N_i	Entry interval
K_{OO}	$[-1/16,\ 1/16]$	$[-1,1]$
K_{OA}	$[-117/256,\ -85/256]$	$[-1,1]$
K_{AB}	$[-1/16,\ 1/16]$	$[-1,1]$
K_{BO}	$[151/512,\ 215/512]$	$[-1,1]$
K_{BA}	$[-1/16,\ 1/16]$	$[-1,1]$

Table 3: The five disjoint source strips.

4 Finite covering conditions

Let $(\mathsf{U}_j, \mathsf{S}_j) = \psi_j^{-1} \circ P$ be target coordinates. For each source strip K_{ij}, the following sufficient conditions are checked on the entire indicated sets:

$$-1 < \mathsf{S}_j(\psi_i(u,s)) < 1 \qquad (a_{ij} \le u \le b_{ij},\ -1 \le s \le 1), \tag{C1}$$

$$\mathsf{U}_j(\psi_i(a_{ij},s)) < -1, \qquad \mathsf{U}_j(\psi_i(b_{ij},s)) > 1 \qquad (-1 \le s \le 1). \tag{C2}$$

No point sampling is substituted for these quantified conditions.

Lemma 2. *Conditions* (C1)–(C2)*, together with continuity of P, imply $K_{ij} \overset{P}{\Longrightarrow} N_j$. They also imply $K_{ij} \overset{P}{\Longrightarrow} K_{jk}$ for every listed strip $K_{jk} \subset N_j$.*

Proof. In normalized source coordinates let $F = (F_u, F_s)$ be the target-coordinate map. The homotopy $((1-t)F_u + 2tu, (1-t)F_s)$ has no intersection with the target entry boundary. On the two source exit edges its first component stays, respectively, below -1 and above 1. At $t = 1$ the exit map is $u \mapsto 2u$, of degree one. This is a covering relation in the sense of [6].

For a target strip with exit interval $[a, b] \subset (-1, 1)$, replace the first target coordinate by $(\mathsf{U}_j - (a+b)/2)/((b-a)/2)$. A value below -1 remains below the new left exit, and a value above 1 remains above the new right exit. The entry coordinate is unchanged. The same homotopy argument therefore applies. □

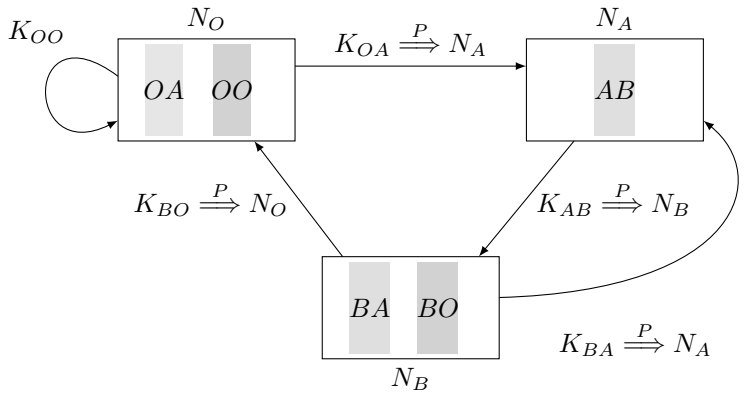


Figure 1: Schematic section boxes and their source strips; the geometry is not to scale. An arrow denotes a covering from its labelled strip to the full target box, hence to each successor strip. It does not assert a covering from an entire parent box.

5 Two interval validations

Both computations use the exact input data above and the same definition of first return. The first implements its own interval propagation and return enclosure. The second uses CAPD's interval integrator and Poincaré map, with a separately written model, initial-set construction, chart, tube audit and acceptance predicates. Its recorded construction excluded the first implementation's source and results (see `VERIFICATION-RECORD.md`), but used the same frozen equations, boxes and strips. Errors in these shared inputs can affect both computations. The two arithmetic choices, orders and subdivisions provide implementation diversity, not independent selection of the geometric claim.

5.1 Initial graph, existence and return isolation

For each parameter cell, the full graph (6) is enclosed. Its nonlinear component uses the affine part of $g(w) = \sqrt{1-w^2}$ and a bound on $g''(w) = -(1-w^2)^{-3/2}$. The common dependence on (u, s) is retained in a correlated set representation. Off-null points may be present in the outer set, but (5) ensures that the intended initial graph follows the null dynamics.

Each implementation checks the integration tubes against the strict inequalities $r_\pm > 1/5$ and $x^2 + z^2 < 64$. The initial upward departure is excluded as a return, intervening crossings are distinguished by direction, and the next upward crossing is isolated with strictly positive v_z. In the binary64 computation, a sequence of sign states records departure, the positive side, a strictly downward crossing, the negative side and the strictly upward return window. On either side, a step is admitted only when its tube avoids $z = 0$ or its velocity sign excludes a new zero. Each crossing window has a fixed nonzero sign of v_z and opposite endpoint signs. This proves that no earlier upward return was skipped. CAPD's oriented Poincaré routine excludes the initial zero and locates the next minus-to-plus crossing; an audit inside each integration step rejects every section-intersecting tube for which v_z can vanish. The recorded downward crossing and positive return time exclude the initial departure.

For any initial point, departure and the isolated crossings are transverse; the intervening closed orbit segments are separated from $z = 0$. These properties and the strict spatial margins persist under sufficiently small changes of the initial point. Smooth dependence of the flow and the implicit-function theorem at the final crossing therefore give continuity of the first return on each closed strip. Only this continuity is used in the covering argument.

5.2 Binary64 implementation

The first implementation uses Taylor order ten. Each basic arithmetic or square-root result, evaluated in nearest rounding, is widened by one representable number in each outward direction using `nextafter`. The compiler options are `-O2 -ffp-contract=off -fno-fast-math`; the trust base includes correctly rounded binary64 operations and square root. The base time step is $h = 1/16$. Parameter rectangles are first split to prescribed uniform width thresholds, then bisected further when an enclosure or covering test remains unresolved. A strict Picard enclosure proves existence on each time step; Taylor remainder bounds enclose both the flow and its variational matrix. Sets are represented as a centre, two correlated parameter columns, and a moving four-dimensional error box. Approximate QR bases are permitted because an interval enclosure of the inverse of the actual stored basis is used in the inclusion update.

At the common time just before the return window, let the state be X_0 and its section coordinate be $z_0 < 0$. The positive lower bound for v_z allows integration with respect to z. With

$$\mathcal{L} \supset \left(\frac{v_x}{v_z}, \frac{v_x'}{v_z} \right) \quad \text{on the whole crossing tube,}$$

the return image satisfies

$$(x, v_x)_{\mathrm{ret}} \in (x, v_x)_0 - \ell_0 z_0 + (\ell_0 - \mathcal{L}) z_0$$

for an arbitrary point vector ℓ_0. Evaluating the first linear expression in the correlated representation preserves useful cancellation. This contains the images at the initial-condition-dependent hitting times, not merely the state at a single common time.

The saved certificate has 768 accepted leaves across all fifteen conditions. A separate exact-rational checker visits all 1521 nodes of the binary partition trees and checks the recorded domain and crossing bounds and strict target inequalities at every leaf. It audits saved bounds and does not reintegrate the ODE.

5.3 CAPD/MPFR implementation

The second implementation uses CAPD 6.1 [7], 160-bit MPFR intervals, Taylor order 24 and the `MpC1Rect2Set` representation. Decimal inputs are read with directed MPFR rounding. The interval Poincaré map uses the upward orientation and an affine target-coordinate evaluation at an interval-enclosed variable return time. An additional tube audit inside the integration checks the spatial boundary margins and rejects every section-intersecting tube whose normal velocity contains zero. Positive return time, positive return velocity and an audited downward crossing are also required.

All 2381 accepted records use the affine return enclosure in `return_check.cpp`, with condition labels C1, L and R. The exploratory mean-value variant M1 contributes no accepted record. Some accepted cells overlap or were retained from earlier subdivisions. The saved-record checker is `evidence/independent-capd/ab-extension/check_coverage.py`. It validates rational cell indices in the unit parameter square, sweeps their first-coordinate endpoints, merges overlapping second-coordinate intervals in each slab, and sums exact rational areas. No clipping is needed because the indices already constrain every cell to the square. It checks the exit edges by interval unions. A finite union of closed rectangles with full area covers the whole closed square: a nonempty relative-open complement would have positive area. Thus the unions cover each complete strip and both exit edges. This audit uses saved records, without a new ODE integration. Unresolved parents are not accepted as evidence. The numerical decision is made by native interval comparisons; printed endpoints and derived margins are diagnostic values, checked separately.

Covering	Binary64 cells (C1; L; R)	CAPD cells (C1; L; R)	Result
OO	$64; 16; 16$	$128; 16; 16$	C1, C2 pass
OA	$64; 16; 16$	$256; 16; 16$	C1, C2 pass
AB	$256; 64; 64$	$1330; 67; 72$	C1, C2 pass
BO	$64; 16; 16$	$272; 16; 16$	C1, C2 pass
BA	$64; 16; 16$	$128; 16; 16$	C1, C2 pass

Table 4: Whole-set covering checks. Binary64 counts are partition leaves; CAPD counts are distinct accepted parameter cells and need not form a disjoint partition. The two counts are not a performance comparison.

	Binary64 conservative displayed bounds		CAPD diagnostic minima (approx.)	
Covering	C1 margin	C2 margin	C1 margin	C2 margin
OO	0.977	11.422	0.509499	9.78842
OA	0.402	370.001	0.161531	320.784
AB	0.500	5.818	0.000517844	0.235386
BO	0.339	9.014	0.000747296	3.61244
BA	0.920	301.081	0.317627	135.569

Table 5: C1 margin means $1 - \max |\mathsf{S}|$ for a cell enclosure. C2 margin is the minimum of $-1 - \sup \mathsf{U}$ on left cells and $\inf \mathsf{U} - 1$ on right cells. CAPD values come from `proof-summary.json`. They are diagnostic values, not replacements for strict native interval predicates.

The CAPD source is `evidence/independent-capd/ab-extension/proof-summary.json`. The subdivision retains a cell as soon as its strict predicate passes, without requiring a prescribed margin. Older overlapping cells can remain in the final union. Its small minimum margins therefore depend on enclosure width and subdivision history; they are neither robustness radii nor a measure of disagreement between implementations. The two saved audits agree on exact inputs and all fifteen covering conditions. No pointwise intersection test of their return

enclosures has been performed. The file `evidence/independent-capd/review-audit.json` identifies every minimizing task, original file, row and complete numerical output line.

For scale, the binary64 certificate encloses all return times in the respective rational intervals

ij	OO	OA	AB	BO	BA
τ	$[15.288, 15.296]$	$[15.274, 15.284]$	$[19.070, 19.110]$	$[15.099, 15.114]$	$[15.085, 15.100]$.

These are outward coarsenings of its recorded bounds in affine time λ. For each strip, the aggregate CAPD return-time range is contained in the corresponding displayed interval. This comparison concerns aggregate time ranges, not pointwise intersection of the two return-image enclosures. Agreement of coverings does not require equality or nesting of enclosures.

6 From coverings to entropy

Order the strips as OO, OA, AB, BO, BA. Lemma 2 gives the transition matrix

$$B = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}. \tag{8}$$

It is the edge matrix of $A = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{smallmatrix}\right)$. The disjoint objects needed for symbolic coding are the five strips, rather than three parent boxes with two incompatible whole-box assertions.

Proof of Theorem 1. The interval certificates give (C1)–(C2), the first-return property and the strict domain and crossing margins. The preceding continuity argument and Lemma 2 prove (1) and every transition in (8).

A closed admissible word for B satisfies the hypotheses of the closed-chain covering theorem [6, Theorem 9]: each strip is a compact h-set with one exit and one entry coordinate, its affine chart extends to a homeomorphism of $\mathbb{R}^2$, P is continuous on each closed source strip, and every forward covering has degree $+1$. No inverse-map hypothesis is used. The theorem gives a point in the prescribed strip interiors fixed by the corresponding iterate of P. Disjointness of the strips makes its itinerary exactly the prescribed periodic word. Its least period need not equal the length of a nonprimitive word. The directed graph of B is strongly connected, so every finite admissible word extends to a periodic one. Let Λ be the closure of the union of such periodic orbits. It is compact and contained in the union of the five strips. Continuity gives $P(\Lambda) \subset \Lambda$; taking convergent subsequences of predecessors on the periodic orbits gives the reverse inclusion.

Because the strips are disjoint compact sets, recording the strip occupied by each forward iterate defines a continuous itinerary map from Λ to the full five-symbol one-sided space. Limits of the periodic itineraries lie in its closed admissible subshift Σ_B^+, so the itinerary of every point of Λ is admissible. Periodic extensions of longer and longer prefixes, followed by a compactness subsequence, realize every admissible one-sided itinerary. This map to Σ_B^+ is onto and intertwines P and the shift. Therefore $h_{\mathrm{top}}(P|_\Lambda) \geq h_{\mathrm{top}}(\sigma|_{\Sigma_B^+}) = \log \rho(B)$. Equivalently, distinct admissible words give orbit segments separated by the positive distance between the strips, and their exponential growth rate is $\rho(B)$.

All graph calculations can be performed over $\mathbb{Q}(\varphi)$ with $\varphi^2 = \varphi + 1$. The strictly positive vector $\xi = (\varphi, 1, \varphi, \varphi, 1)^{\mathsf{T}}$ satisfies $B\xi = \varphi\xi$. The induced weighted maximum norm of B is φ, which gives $\rho(B) \leq \varphi$, while the eigenvalue gives the opposite inequality. The saved integer

calculations also give $\det(\zeta I - B) = \zeta^3(\zeta^2 - \zeta - 1)$. An integer-square-root enclosure supplies rational lower and upper bounds for $\sqrt{5}$. Substitution into the positive-term series

$$\log \varphi = 2 \sum_{k=0}^{\infty} \frac{(\sqrt{5} - 2)^{2k+1}}{2k + 1}$$

and an explicit tail bound, all evaluated by rational arithmetic, yield

$$\log \varphi \in [0.481211825059603, 0.481211825059604].$$

□

7 Reproducibility and limits

The accompanying certificate package contains the paper sources, finite evidence and offline checking tools. Its file inventory is fixed by `MANIFEST.sha256`; citation and archive status are recorded in its README. The packet fixes the exact inputs, numerical sources, accepted records and hashes. Its offline checkers audit all binary partition leaves or exact unions of accepted CAPD cells, and can be run without a new ODE integration. Reproducing the numerical enclosures requires the specified compiler/arithmetic environment for the first implementation or the pinned CAPD/MPFR build for the second. These are different levels of reproduction; checking a saved inequality is not a reimplementation of the underlying interval integrator.

English method descriptions and the expected versus generated checker filenames are listed in `evidence/README.en.md`. The recorded software environment and timing information are summarized in Appendix B.

The numerical trust base includes each interval implementation, elementary arithmetic and functions, compiler and execution environment. Neither the integrators nor the topological argument are formally verified here. The proposed formalization level is L0: the current argument combines executable interval certificates with an external covering theorem, and a Lean statement alone would not certify that interface. The correspondence between the two computations is detailed in `CORRESPONDENCE.md`.

According to its publicly accessible abstract, Pantig [8] develops a conditional symbolic framework for multi-black-hole photon scattering. Our claim is the five explicit coverings and the stated bound per return. A full-text comparison remains pending, so no exhaustive priority assessment is made against that work.

Use of AI. AI tools, including Codex and Claude, were used in implementation, checking records, literature triage and drafting. They are tools, not authors. Submission remains subject to the named author's review and responsibility; this draft has not undergone human peer review.

A A compact optical core

Here is the application of [5, Theorem 2.1] used for context in the introduction. On $M = \mathbb{R}^2 \setminus \{(0, 1), (0, -1)\}$ take the smooth optical metric $g_{\mathrm{opt}} = \mathcal{U}^4(dx^2 + dz^2)$ as the base metric, with potential $V = 0$ and energy $h = 1/2$. The Jacobi metric is then g_{opt}. There are no potential singularities on M, so $\Delta = \varnothing$ and $A(\Delta) = 0$. The domain $\mathcal{D}$ is a disk of radius $R = 8$ with the two open disks of radius $\varepsilon = 1/5$ removed; hence $\chi(\mathcal{D}) = -1$ and $0 > 2\chi(\mathcal{D})$.

For the inward Euclidean unit normal n and its signed Euclidean curvature k_e, the optical geodesic curvature is

$$k_g = \mathcal{U}^{-2}(k_e - 2\partial_n \log \mathcal{U}).$$

On the outer circle, $k_e = 1/R$ and n points towards its centre. Since $r_\pm \geq R - 1$, $\mathcal{U} \geq 1$ and $\partial_r \mathcal{U} \geq -2/(R-1)^2$,

$$\mathcal{U} + 2R\partial_r \mathcal{U} \geq 1 - \frac{4R}{(R-1)^2} = \frac{17}{49} > 0.$$

On either inner circle, n points away from that hole's centre and $k_e = -1/\varepsilon$. The other centre is at least $2 - \varepsilon$ away, and consequently

$$\mathcal{U} + 2\varepsilon\partial_\varepsilon \mathcal{U} \leq 1 - \frac{1}{\varepsilon} + \frac{1}{2-\varepsilon} + \frac{2\varepsilon}{(2-\varepsilon)^2} = -\frac{269}{81} < 0.$$

Thus $k_g > 0$ on all three boundary components, in the inward-normal convention of [5]. The energy and convexity hypotheses are satisfied. The resulting compact positive-entropy set for the optical geodesic flow is a qualitative consequence of that existing theorem. This argument is our application of it, and supplies neither the five strip coverings nor their numerical entropy bound per return.

B Recorded computation environment

Both recorded environments were macOS 26.6.2 on arm64. The binary64 solver used Apple clang 21.0.0 with the following options, recorded in `evidence/hset-check/trials.json`:

```
-std=c++17 -O2 -ffp-contract=off -fno-fast-math
```

The CAPD build used Homebrew GCC 16.2.0, MPFR 4.2.2 and GMP 6.3.0. CAPD 6.1 was fixed at source commit `37057e5899bccaa6cd5701057b7e6ea5e232ef8b`; its source-archive hash and complete compile/link argument arrays are in `environment-record.json` under `evidence/independent-capd/`. The numerical executable used the options

```
-std=c++17 -O2 -frounding-math
-D__USE_NATIVE__ -D__HAVE_MPFR__
```

and linked CAPD, MPFR and GMP. Although the library configuration selects the NATIVE backend and enables multiprecision, this solver uses 160-bit `MpInterval`, not native double intervals. The historical build record specifies the architecture but not the CPU model.

CAPD uses Taylor order 24, a step-selection tolerance of 10^{-20} and a maximum step of 0.2. These select time steps; certified errors come from interval Taylor enclosures rather than from the tolerance itself. The recorded library build took 110.037 seconds. The six saved runs contributing accepted CAPD cells took, respectively, 151.775, 194.846, 1977.723, 198.666, 121.747 and 7251.217 wall seconds, in the order listed in the environment record. These times include unresolved adaptive ancestors and are neither summed CPU times nor a benchmark against binary64. Exact task tuples, failed attempts and individual timings are preserved with the packet; no new integration was performed for this manuscript revision.

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